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Full Legal Name: Arush Duggal
Institution: Lexington High School
Category: Junior
Class/Grade/Year: Grade 10 (Sophomore)
Country: United States of America
Submission Date: September 19, 2026
Registered Email Address: arushduggal1612@gmail.com
Research Topic: NANOSHIELD: A Comparative Evaluation of Electrospun Nanofiber Membranes as Low-Cost Alternatives to Commercial HEPA/ULPA Filtration Media

NANOSHIELD: A Comparative Evaluation of Electrospun Nanofiber Membranes as Low-Cost Alternatives to Commercial HEPA/ULPA Filtration Media

Abstract

Indoor and outdoor air quality pose many public health concerns. With advancing technology, exposure to fine and ultrafine particulate matter has only increased and is linked to serious health effects, including respiratory and cardiovascular harm [1][2]. Commercial HEPA and ULPA filters remain the gold standard for a high efficiency capturing system for capturing harmful particles like fine and ultrafine matter. Still, their high cost and non-biodegradable materials, with an estimated 2.5 million tons of filter waste annually, limit access, specifically for lower-income households and schools [3]. Recent technology shows that electrospun nanofiber membranes made from low-cost, biodegradable polymers can match or even exceed the performance of commercial HEPA-level filtration efficiency at a lower pressure drop [4][5]; however, the majority of this work was confined to a controlled laboratory setting with limited evidence on cost-per-performance against commercial alternatives or practical translation into a buildable device [3][6].

To bridge this gap, this study develops NANOSHIELD, a research and prototyping project that builds and tests electrospun PVA/PLA nanofiber membranes against commercial filters, making a low-cost air purifier prototype constructed around the best-performing material.

Research Questions

This study evaluates whether electrospun nanofiber membranes can be used as a low-cost, well-performing alternative to HEPA/ULPA filters commercially used. Particularly, this study intends to answer the following questions in detail:

1. How effective are electrospun PVA/PLA nanofiber membranes in terms of:
    a. Particle capture effectiveness across PM2.5 and ultrafine, sub 100nm, ranges
    b. Pressure drop with a fixed airflow
    c. Quality factor of the efficiency relative to pressure drop
    d. Cost per unit
2. How effective are commercial filters, in terms of the same four measures, to establish a baseline for comparison?
3. Is there a significant difference between electrospun nanofiber membranes and commercial filters in terms of
    a. Particle capture efficiency
    b. Pressure drop
    c. Quality factor
4. Can a functional, low-cost air purifier be built around an electrospun nanofiber membrane, and how does its real world particle reduction compare to a baseline unit?
5. What is the resulting cost-per-unit performance of the nanofiber-based prototype when compared to a similar sized HEPA based unit?

Research Variables & Hypotheses

To keep a standard format as per academic standards, the experimental layout is defined by the following variables and hypotheses:

Experimental Variables:

- Independent variables:
    - Filter type - Experimental PVA, Experimental PLA, Commercial MERV 8, Commercial MERV 13, True HEPA

    - Electrospinning voltage
    - Polymer concentration
- Dependent Variables:
    - Particle Fractional Efficiency
    - Static Pressure drop
    - Calculated Quality Factor
    - Localized room particle decay rate
- Controlled Variables:
    - Volumetric airflow rate
    - Upstream challenge particle mass concentration
    - Sensor sampling frequency

Hypotheses:

1. If the polymer concentration and the distance from the collector are fine-tuned during electrospinning to achieve an average fiber distance of 150-300 nm, then the membranes will eithe match or exceed true HEPA efficiency (>99.97%) while demonstrating a lower static pressure drop. Sub-micron fiber matrices leverage slip-flow mechanics, which allow gas molecules to slip along the fiber surface, minimizing air resistance while maximizing interceptive and diffusional particle capture mechanics [4][6].

2. If a composite nanofiber matrix is integrated into a custom 3D-printed appliance utilizing tight gasket compression, then the resulting prototype will deliver comparable clean air delivery performance at less than 30% of the manufacturing and material costs of retail appliances, providing a mathematically viable avenue for domestic deployment.

Literature Review

Exposure to fine and ultrafine particulate matter is a well-documented health concern, driving continued demand for accessible, high-efficiency air filtration [1][2]. Traditional HEPA filters, while effective, aren't biodegradable and continue to add significant waste upon disposal. Researchers have pointed towards more sustainable alternatives with better performance.

Biabani et al. [4] reviewed nanofiber modification strategies for particle filtration and discovered that the majority of studies rely on self-charging materials to boost capture efficiency, with little attention paid to cost or equipment accessibility. Keyvani et al. [5] created bio-based, flame-retardant PVA nanofiber membranes and found that although raw efficiency was slightly below a commercial HEPA filter, the membranes' much lower pressure drop gave them a much higher quality factor. A related systematic review by Keyvani et al. [6] evaluated biodegradable electrospun membranes specifically for submicron particle capture, identifying fiber diameter and membrane thickness as the dominant performance variables.

Harikrishnan et al. [3] developed PLA nanofibers with a metal-organic framework (ZIF-67). Achieving 99.22-99.36% PM2.5 removal efficiency while adsorbing toluene, which is a common volatile organic compound used indoors. Parasuraman et al. [7] formed a reusable polyetherimide nanofiber membrane achieving a 99.971% PM2.5 removal at a strikingly low 71 Pa pressure drop, outperforming both an N95 respirator and a commercial HEPA filter under identical test conditions.

Taken together, the literature supports an integrated model: nanofiber media can match or exceed commercial HEPA performance in controlled lab settings, using biodegradable, relatively low-cost base polymers. At the same time, prior work highlights the real-world questions this study must address—practical fabrication accessibility, cost-per-performance against commercial media, and translation into a working device—which inform this study's design and evaluation priorities [4][6][7].

Proposed Methodology

The method is a quantitative, comparative-experimental approach.

1. Baseline Filter Evaluation:
   A sealed, airtight acrylic test housing will be constructed. It will feature a controllable variable-speed blower fan to maintain a constant face velocity across the media samples. Particulate monitoring will be performed using calibrated Plantower PMS5003 and PMS7003 optical sensors positioned upstream and downstream of the filter holder.
   Commercial grade filter samples (MERV 8, MERV 13, and fiberglass true HEPA) will be trimmed to standard dimensions (10 x 10 cm) and tested with a fixed airflow of 2.0 L/min. The sensors will document real world particle count reductions across specific size ranges (PM1.0, PM2.5, and PM10). A digital differential manometer will record the static pressure drop in Pascals across the filter face.

2. Nanofiber Synthesis Protocol
   In collaboration with a university materials science laboratory, experimental PVA and PLA membranes will be fabricated via needle-based electrospinning. Polymeric solutions will be prepared by dissolving PVA in deionized water (10 wt%) and PLA in a chloroform/dimethylformamide mixture (8 wt%). The parameters will be monitored under laboratory supervision: voltage will vary between 18–22 kV, the feed rate will be held at 0.5 mL/h, and the tip-to-collector distance will be maintained between 15–20 cm. The fibers will be collected directly onto an open-mesh scrim substrate. A secondary composite variant will be made by spraying activated carbon particles onto the collecting fibers to evaluate simultaneous particulate and gas-phase VOC reduction.

3. Mathematical Evaluation & Quality Factor Calculations
   Using the custom baseline test rig, the experimental nanofiber membranes will be evaluated under identical flow velocity conditions. The fractional efficiency (η) of each sample is calculated using the upstream (N_up) and downstream (N_down) particle counts:

   $$\eta = 1 - (N_{down} / N_{up})$$

   To determine the true performance optimization balancing particle capture and energy consumption, the formal Quality Factor (Qf) will be calculated for every commercial and experimental medium:

   $$Qf = -\ln(1 - \eta) / \Delta P$$

   Where:

   η = the fractional collection efficiency for PM2.5

   ΔP = the static pressure drop across the filter face (Pa)

   ln = the natural logarithm

4. Cost-Performance & Economic Analysis
   To establish practical real-world feasibility, material and manufacturing costs will be tracked. A Normalized Cost-Performance Metric (C_perf) will be calculated by dividing each filter's Quality Factor by its total raw material cost per unit area:

   $$C_{perf} = Qf / (\text{Cost per } m^2)$$

   A one-way Analysis of Variance (ANOVA) will be executed via a Python script to determine if the variations in Qf and C_perf between the electrospun membranes and commercial baselines are statistically significant ($p < 0.05$).

5. Prototype Construction and Room Validation Testing
   The highest-performing experimental membrane will be integrated into a functional air purifier chassis engineered via Onshape and constructed using an FDM 3D printer. The housing will incorporate a tight gasket-compression layout to ensure no air bypasses the filter media.
   Validation testing will be conducted in a sealed 1 m^3 chamber challenged with localized incense smoke. The

internal particle concentration decay over time will be recorded to calculate the clean air delivery rate, which will be directly compared against a retail commercial HEPA air purifier.

Project Timeline:

The project will span approximately 6 months and proceed through the following phases:

Month 1:	Build the baseline test rig and acquire commercial filter samples.
Months 2–3:	Fabricate nanofibers at the host laboratory.
Months 4–5:	Conduct comparative testing, perform statistical analysis, and construct the prototype.
Month 6:	Execute validation testing, finalize results, and prepare the full report.

Predicted Results:

The study is envisioned to show whether or not electrospun PVA/PLA nanofiber membranes can serve as a viable alternative to HEPA/ULPA filtration. It will create a relative dataset traversing efficiency pressure drop, quality factor, and cost per performance, across commercial and experimental filters. This will result in a working low-cost air purifier prototype built around the best-performing material. The results will be used to decide whether recent nanofiber research can be practically translated into, working, accessible real-world filtration devices.

Potential Limitation:

Due to changes in polymer batch quality, humidity during electrospinning, and limited access to sub 100nm particle counting instruments outside of the host laboratory, the study may experience some variability in nanofiber performance and constrained precision in the ultrafine particulate matter range.

Conclusion

In conclusion, NANOSHIELD, gives a practical, evidence based finding of low cost electrospun PVA/PLA nanofiber membranes as a viable alternative to HEPA/ULPA filters found commercially. By using both rigorous comparative testing and the construction of a working prototype, the study bridges the gap between laboratory nanofiber research, and real world application. Eventually, this work aims to support clean air by sustaining broader access to effective, affordable filtration technology, by targeting households and schools where standard commercial filtration systems remain inefficient in both performance and cost

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